

Summer Review Plan

First-Year Graduate Physics Core

Classical & Statistical Mechanics · Quantum Mechanics · Electrodynamics

Window: 1 June 2026 – 30 August 2026 · 13 weeks

Budget: 10 hours/week · 130 hours total

Split: ~3.3 hr per track · ~2 hr reading + ~1.3 hr problems

Track	Primary text	Secondary
Classical Mechanics	Lemos, Analytical Mechanics (2018)	Taylor; Goldstein
Statistical Mechanics	Schroeder (core) + Pathria (depth)	—
Quantum Mechanics	Merzbacher, QM 3rd ed.	—
Electrodynamics	Zangwill, Modern Electrodynamics	—

How this plan is built

The plan treats **CM + SM as a single track** (matching a first-year sequence that covers classical then statistical mechanics), and gives **QM** and **E&M** their own tracks. Each track gets roughly 3.3 hours per week. Within a 3.3-hour block, the default split is ~2 hours reading and ~1.3 hours problem-solving.

Weekly rhythm (a template that fits in 10 hrs)

Day	Activity	Hours
Mon	QM — read assigned sections	1.5
Tue	QM — problems (2)	1.5
Wed	CM/SM — read	2.0
Thu	CM/SM — problems (2)	1.3
Fri	E&M — read	2.0
Sat	E&M — problems (2) + 15-min Sunday review of all 3	1.7
Sun	Light: review notes, scan next week	(buffer)

You can swap days freely — the only hard rule is to **not skip problems**, since passive reading will not build the recall speed you actually need.

Problem-selection philosophy

Two carefully chosen problems per track per week (≈ 6 problems/week, ≈ 78 across the summer). Pick one that is **computational** (practice a technique) and one that is **conceptual or synthetic** (ties the chapter together or connects to a qual-style question). Starred problems in the primary texts are usually the right ones; if the chapter has no stars, pick a mid-range problem (not the first trivial one, not the open-ended research one).

If you finish both problems in under 1 hour: add one more from the same chapter. **If you cannot finish one in 1.5 hours:** stop, write what blocked you, move on. Come back to it next week.

Personalized reweighting (from your self-assessment)

You flagged the following as weak or not-yet-covered. The weekly detail blocks below reflect these priorities — priority weeks are marked with ★ in the master calendar and get an extra problem and deeper reading. Other weeks stay at review pace.

Track	Your weak / new topics	How the plan responds
CM	Rotations · Small oscillations / normal modes	★ Weeks 3 and 5 are expanded: +1 problem each, slower reading pace, extra reference to Goldstein Ch 4–5.
SM	All three ensembles · Phase transitions · Mean field theory · Brownian motion / random walks	Weeks 9–13 restructured. Canonical (wk 9), grand canonical (wk 10), phase transitions + Landau mean field (wk 12), Brownian motion + random walks (wk 13). BE/FD still covered in wk 11.
QM	3D potential wells · Spin	★ Weeks 9 and 11 are expanded: +1 problem each, deeper reading into Merzbacher Ch 12 and Ch 16.
QM	Path integrals · CFT (never done)	These are new learning, not review. Path integrals get a brief intro via Merzbacher §15.3 in week 12 (stretch). CFT (classical or conformal) is deferred — see the closing note on p.11.
E&M;	(no weak areas reported)	Unchanged — all 13 weeks at review pace.

Quick self-diagnostic grid (for reference)

Since part of the goal is to shore up weak areas, spend ~30 min in the last week of May rating your comfort (1–5) with each of the topics below. On any topic you rate 1–2, add ~20% more time (at the cost of reading on a topic you rated 4–5 elsewhere).

CM/SM	QM	E&M
Lagrangian with constraints	1D scattering / tunneling	Boundary-value problems
Noether / conservation laws	Harmonic oscillator (operator)	Method of images
Rigid body / Euler eqns	Angular momentum algebra	Multipole expansion
Small oscillations	Hydrogen atom	Magnetostatics w/ vector pot.
Poisson brackets / can. transf.	Spin, Pauli matrices	Waves in conductors
Microcanonical / canonical	Addition of angular mom.	Retarded potentials
Grand canonical	Time-indep. perturbation	Larmor / dipole radiation
Bose/Fermi gas	Time-dep. perturbation	4-vector EM (relativity)

Master calendar — 13 weeks at a glance

Wk	Dates	CM / SM	QM (Merzbacher)	E&M (Zangwill)
1	Jun 1–7	Lemos Ch 1 §1.1–1.5 — Lagrangian basics	Ch 1 + Ch 2 — wave packets	Ch 1 §1.1–1.6 + Ch 2 — math + Maxwell intro
2	Jun 8–14	Lemos §1.6–1.7, Ch 2 — central force, Noether	Ch 3 + Ch 4 §1–5 — Schrödinger, principles	Ch 3 §3.1–3.7 — electrostatics core
★3	Jun 15–21	Lemos Ch 3 — rotations, Euler angles (priority)	Ch 5 + Ch 4 §6–7 — harmonic osc., gauge	Ch 4 §4.1–4.6 — electric multipoles
4	Jun 22–28	Lemos Ch 4 — rigid body dynamics	Ch 6 — 1D sectionally const. potentials	Ch 7 §7.1–7.7 — Laplace, sep. of vars.
★5	Jun 29–Jul 5	Lemos Ch 5 — small oscillations (priority)	Ch 7 — WKB	Ch 8 §8.1–8.5 — images, Green functions
6	Jul 6–12	Lemos Ch 7 + Ch 8 §1,4,5 — Hamiltonian, Poisson	Ch 8 §1–4 — variational + pert. intro	Ch 10 §10.1–10.5 — magnetostatics
7	Jul 13–19	Schroeder Ch 1–2 — energy, 2nd law	Ch 9 + Ch 10 §1–5 — vector spaces	Ch 11 + Ch 12 — magnetic multipoles, forces
★8	Jul 20–26	Schroeder Ch 3 + Pathria §1.1–1.4 — microcanonical focus	Ch 10 §6–7 + Ch 11 §1–4 — ang. mom.	Ch 14 §1–10 — dynamic/quasistatic fields
★9	Jul 27–Aug 2	Pathria Ch 3 + Schroeder Ch 5 — canonical ensemble deep dive	Ch 12 — central potentials, hydrogen (priority)	Ch 15 §1–5 + Ch 16 §1–4 — conservation, waves
★10	Aug 3–9	Pathria Ch 4 + Schroeder Ch 7.1 — grand canonical ensemble	Ch 13 §1–5 — scattering, Born, phase shifts	Ch 17 §1–4 — waves in matter, R&T;
★11	Aug 10–16	Schroeder Ch 7 + Pathria Ch 6–8 — BE/FD gases	Ch 16 §1–5 — spin, Pauli matrices (priority)	Ch 20 §1–5 — retarded potentials, radiation
★12	Aug 17–23	Pathria Ch 11–12 — phase transitions + Landau mean field	Ch 17 + Ch 18 + §15.3 path-integral intro (stretch)	Ch 20 §6–8 + Ch 21 §1–4 — multipole radn., scattering
★13	Aug 24–30	Pathria Ch 15 — Brownian motion / random walks + synth.	Ch 19 §1–5 — time-dep. perturbation + synth.	Ch 22 §1–7 — special relativity in E&M;

Reading volumes were sized to ~20–30 pages of dense primary text per week per track. Section numbers follow the primary edition of each book (Lemos 2018, Zangwill 2013, Merzbacher 3rd ed., Schroeder 2000, Pathria 3rd ed.).

Track 1 — Classical & Statistical Mechanics

Weeks 1–6 are CM (Lemos). Weeks 7–13 are SM (Schroeder + Pathria). Each week uses ~3.3 hours. Keep Taylor and Goldstein on the desk for when Lemos is too terse: Taylor for intuition and worked examples (especially Ch 7, 11, 13), Goldstein for canonical transformations and the Hamilton–Jacobi chapter.

Part A: Classical Mechanics (Lemos, weeks 1–6)

Week 1 (Jun 1–7) · Lagrangian foundations

Read (~2 hr): Lemos §1.1 Newtonian review · §1.2 constraints · §1.3 virtual displacements / d'Alembert · §1.4 generalised coords and Lagrange's eqns · §1.5 applications. Cross-reference: Taylor Ch 7.

Problems (~1.3 hr): Lemos Ch 1, pick **one** constrained-system problem (e.g., bead on a rotating wire or double pendulum) + **one** problem that requires writing the Lagrangian in generalised coordinates. Typical choices: 1.3, 1.6, 1.9, 1.11.

Focus: Get fluent writing $L = T - V$ in arbitrary coordinates; this is the muscle you lean on all year.

Week 2 (Jun 8–14) · Central forces, variational principle, Noether

Read (~2 hr): Lemos §1.6 generalised potentials/dissipation · §1.7 central forces + Bertrand's thm · Ch 2 in full: variational principle, non-holonomic case, Noether.

Problems (~1.3 hr): **One** Kepler/central-force problem from Ch 1 (orbits, turning points) + **one** Noether-style problem from Ch 2 (construct a conserved quantity from a symmetry). Good picks: 1.14 or 1.16; 2.5 or 2.8.

Focus: Conservation laws $\leftrightarrow$ symmetries is the single most transferable idea in the course.

★ Week 3 (Jun 15–21) — priority · Rotations, Euler angles, rotating frames

Read (~2 hr): Lemos Ch 3 §3.1–3.6 — orthogonal transformations, Euler angles, infinitesimal rotations, non-inertial frames (Coriolis, centrifugal). **Read twice.** Pre-read Taylor Ch 9 first for intuition; then Lemos for the formal structure; then Goldstein Ch 4 for the matrix/tensor formalism.

Problems (~1.3 hr): **Three** problems this week (+1 over baseline): (a) one Euler-angle transformation problem (rotate a vector or find composed rotations), (b) one non-inertial-frame problem (Foucault pendulum or projectile Coriolis deflection), (c) one infinitesimal-generator problem (show $[\mathbf{L}, \mathbf{L}_k] = \epsilon_{ijk} \mathbf{L}_j$ classically).

Focus: Priority week. Rotations is one of the two CM topics you flagged as weak — this is where you invest the extra hour. Aim for real comfort with active/passive conventions and the difference between body-fixed and space-fixed frames.

Week 4 (Jun 22–28) · Rigid-body dynamics

Read (~2 hr): Lemos Ch 4 §4.1–4.9 — inertia tensor, principal axes, Euler's eqns, symmetric top.

Problems (~1.3 hr): **One** moment-of-inertia / diagonalisation problem + **one** symmetric-top or free-rotation problem. Choose problems from Ch 4 that involve computing the tensor from scratch and one that asks about torque-free motion.

Focus: Symmetric top is a qual classic. If short on time, skip §4.7 (rolling coin).

★ Week 5 (Jun 29–Jul 5) — priority · Small oscillations and normal modes

Read (~2 hr): Lemos Ch 5 §5.1, 5.3–5.6 — 1D case, stationary motion + small oscillations, general case, normal modes, normal coordinates. Supplement with Taylor Ch 11 for worked examples. If you have time, look at Goldstein Ch 6 for the molecular-vibrations perspective.

Problems (~1.3 hr): **Three** problems this week (+1 over baseline): (a) 2-DOF coupled oscillators — find normal modes and frequencies analytically; (b) a 3-DOF symmetric system (triatomic molecule or triple pendulum) — use symmetry to guess modes, then verify; (c) a damped- and/or driven-mode problem to connect to resonance.

Focus: Priority week. 'Modes of motion' is the other CM topic you flagged. Goal: write M and K matrices by inspection from any Lagrangian, then diagonalise without stalling.

Week 6 (Jul 6–12) · Hamiltonian formalism, Poisson brackets

Read (~2 hr): Lemos Ch 7 §7.1–7.3 (canonical eqns, symmetries, virial) + Ch 8 §8.1 (generating functions), §8.4 (Poisson brackets), §8.5 (infinitesimal canonical transformations). Skim §8.7–8.9.

Problems (~1.3 hr): **One** canonical-transformation problem (verify a transformation is canonical via generating function) + **one** Poisson-bracket computation (angular momentum algebra or time evolution of a constant of motion).

Focus: Hamilton–Jacobi (Lemos Ch 9) is deferred — revisit it next year if needed.

Part B: Statistical Mechanics (Schroeder + Pathria, weeks 7–13)

Use Schroeder for intuition and physical arguments, Pathria for the formal ensemble machinery. Read Schroeder first in a given week, then lift Pathria sections that give the rigorous version. If the two disagree on convention, Pathria is authoritative for grad-level work.

Week 7 (Jul 13–19) · Energy and the second law

Read (~2 hr): Schroeder Ch 1 (energy in thermal physics: equipartition, heat, work) and Ch 2 (combinatorics, multiplicity, 2nd law qualitatively).

Problems (~1.3 hr): Two Schroeder problems: one computational (multiplicity of a two-state system or Einstein solid) and one conceptual (why entropy increases). Typical: 2.18 and 2.22.

Focus: Schroeder's combinatorial entropy argument is the cleanest intuition you'll find anywhere.

★ Week 8 (Jul 20–26) — priority · Microcanonical ensemble (entropy, temperature)

Read (~2 hr): Schroeder Ch 3 (interactions and implications — temperature from multiplicity, entropy of ideal gas, paramagnet) + Pathria Ch 2 §2.1–2.5 (microcanonical ensemble: phase-space volume, density of states, entropy from Ω).

Problems (~1.3 hr): Three problems (+1 over baseline): (a) Schroeder 3.x temperature-from-multiplicity (two Einstein solids in thermal contact); (b) derive Sackur–Tetrode for the ideal gas from $\Omega(E)$; (c) one Pathria 2.x problem — compute $\Omega(E)$ for a harmonic-oscillator ensemble or paramagnet.

Focus: Priority week — you flagged all three ensembles as weak. This is the foundation; get $\Omega(E) \rightarrow S \rightarrow T$ automatic before moving on.

★ Week 9 (Jul 27–Aug 2) — priority · Canonical ensemble (deep dive)

Read (~2 hr): Pathria Ch 3 §3.1–3.7 (canonical ensemble, partition function, thermodynamic connection, fluctuations in energy, equipartition, virial) + Schroeder Ch 6 §6.1–6.3 (Boltzmann factor, partition function). Compare with Schroeder Ch 5 §5.1 (Helmholtz free energy) for intuition.

Problems (~1.3 hr): Three problems (+1 over baseline): (a) compute Z and derive all thermodynamics for a two-level system or paramagnet in an external field; (b) do the same for a classical rotor or diatomic molecule (to see where translational / rotational / vibrational Z 's factor); (c) a fluctuation problem — compute $\langle(\Delta E)^2\rangle$ and relate it to C_V .

Focus: Priority week. Z is the central object of SM — don't leave until $\langle E \rangle = -\partial \ln Z / \partial \beta$ and $\langle(\Delta E)^2\rangle = k_B T^2 C_V$ are both automatic.

★ Week 10 (Aug 3–9) — priority · Grand canonical ensemble

Read (~2 hr): Pathria Ch 4 §4.1–4.5 (grand canonical ensemble: grand partition function Ξ , chemical potential, fluctuations in particle number, equilibrium between phases) + Schroeder §7.1 (Gibbs factor, chemical potential intuition).

Problems (~1.3 hr): Three problems (+1 over baseline): (a) derive $\langle N \rangle$ and $\langle(\Delta N)^2\rangle$ for an ideal gas from Ξ ; (b) a Langmuir adsorption problem (surface sites, grand canonical treatment); (c) verify that $\mu = (\partial F / \partial N)_{T,V}$ agrees between canonical and grand canonical.

Focus: Priority week. Grand canonical is the gateway to BE/FD next week — build the habit of switching $(T, V, N) \leftrightarrow (T, V, \mu)$ by Legendre transform.

★ Week 11 (Aug 10–16) — priority · Bose–Einstein and Fermi–Dirac statistics

Read (~2 hr): Schroeder Ch 7 §7.2–7.4 (quantum statistics, degenerate Fermi gas, BEC intro) + Pathria Ch 6 §6.1–6.3 (formulation of quantum stats) + Ch 7 §7.1 (photon gas) + Ch 8 §8.1 (ideal Fermi gas thermodynamics, Sommerfeld expansion setup).

Problems (~1.3 hr): Two problems: (a) degenerate Fermi gas — electron gas in a metal or a white-dwarf estimate (find E_F , P at $T = 0$); (b) black-body radiation / photon gas — derive Stefan–Boltzmann from Ξ .

Focus: Applies the grand canonical machinery from week 10. Know Sommerfeld expansion as a technique, not a memorised formula.

★ Week 12 (Aug 17–23) — priority · Phase transitions + mean field theory

Read (~2 hr): Pathria Ch 11 §11.1–11.5 (order parameter, broken symmetry, Landau theory, Ising model intro) + Ch 12 §12.1–12.2 (mean-field treatment of the Ising model, critical exponents from mean field). Supplement with Schroeder §5.3 (Clausius–Clapeyron) for first-order transition intuition.

Problems (~1.3 hr): Three problems (+1 over baseline): (a) derive mean-field T_c for the Ising model and the magnetisation $m(T)$ near T_c ; (b) compute critical exponents $\alpha, \beta, \gamma, \delta$ in the Landau expansion; (c) a Clausius–Clapeyron coexistence problem (vapour pressure / latent heat).

Focus: Priority week. You flagged phase transitions + mean field as weak. Focus on the logic of Landau theory (expand F in the order parameter, respecting symmetry) more than the specific Ising computation.

★ **Week 13 (Aug 24–30) — priority · Brownian motion, random walks + synthesis**

Read (~2 hr): Pathria Ch 15 §15.1–15.4 (Brownian motion: Langevin equation, fluctuation–dissipation, diffusion, random walks, first-passage). If Pathria feels dense, the Wikipedia derivation of the Langevin equation plus Chandrasekhar's classic review (RMP 1943) are excellent supplements.

Problems (~1.3 hr): Two problems: (a) solve the Langevin equation for $\langle x^2(t) \rangle$ of a Brownian particle, showing ballistic $\rightarrow$ diffusive crossover; (b) a 1D random walk problem — compute the diffusion coefficient and mean first-passage time.

Focus: Priority week — you flagged this as barely covered. Close with a 30-min synthesis: write a 1-page cheat sheet (ensembles, free energies, derivatives, fluctuation–dissipation).

Track 2 — Quantum Mechanics

All weeks use Merzbacher, Quantum Mechanics (3rd ed., 1998). Merzbacher is denser than Griffiths; budget more reading time in formalism chapters (Ch 9, 10, 14). If a chapter feels opaque, skim Griffiths or Sakurai for the same topic before returning.

Week 1 (Jun 1–7) · Wave mechanics + wave packets

Read (~2 hr): Merzbacher Ch 1 (intro) + Ch 2 (wave packets, uncertainty relations, motion of a packet). Light reading.

Problems (~1.3 hr): **One** Gaussian wave-packet problem (compute $\langle x \rangle$, Δx , Δp , and their time evolution) + **one** superposition problem from Ch 2 problems.

Focus: Re-derive the Gaussian packet from scratch — this is the cleanest wave-mechanical example.

Week 2 (Jun 8–14) · Schrödinger eq. + principles

Read (~2 hr): Merzbacher Ch 3 (wave eq., probabilities, operators, virial thm) + Ch 4 §4.1–4.5 (Hermitian operators, completeness, continuous spectrum, displacement operator).

Problems (~1.3 hr): **One** commutator / operator-algebra problem (Ch 3) + **one** eigenfunction / completeness problem (Ch 4).

Focus: The habit to build: always write expectation values as $\langle \psi | \hat{O} | \psi \rangle$, never integrals by eye.

Week 3 (Jun 15–21) · Harmonic oscillator + gauge invariance

Read (~2 hr): Merzbacher Ch 5 (full, short) — oscillator by series method + ladder operators. Then Ch 4 §4.6–4.7 — charged particle in EM field, Galilean transformation.

Problems (~1.3 hr): **One** SHO ladder-operator problem (compute $\langle x^2 \rangle$ in n th state, or matrix element $\langle n | x | n+1 \rangle$) + **one** gauge-transformation problem.

Focus: Master the raising/lowering algebra before Ch 10 revisits it.

Week 4 (Jun 22–28) · 1D sectionally constant potentials

Read (~2 hr): Merzbacher Ch 6 — step, barrier, square well, tunnelling, symmetries.

Problems (~1.3 hr): **One** transmission/reflection problem (compute T , R for a barrier or step) + **one** bound-state problem (square well, count states and find energies graphically).

Focus: These problems appear on quals every year — aim for speed, not novelty.

Week 5 (Jun 29–Jul 5) · WKB approximation

Read (~2 hr): Merzbacher Ch 7 — method, connection formulas, bound states, barrier tunneling.

Problems (~1.3 hr): **One** WKB bound-state problem (quantisation condition for a non-trivial potential) + **one** tunneling probability via WKB (e.g., α -decay toy model).

Focus: Connection formulas are painful — derive them once, then use the result.

Week 6 (Jul 6–12) · Variational method + intro perturbation

Read (~2 hr): Merzbacher Ch 8 §8.1–8.4 — calculus of variations in QM, Rayleigh–Ritz, elementary time-independent perturbation theory, non-orthogonal basis.

Problems (~1.3 hr): **One** variational ground-state estimate (hydrogen or helium, a simple trial function) + **one** first-order perturbation problem from Ch 8.

Focus: Variational method is cheap power — pick an easy trial ψ , get a surprisingly good bound.

Week 7 (Jul 13–19) · Hilbert space and operator eigenvalue problem

Read (~2 hr): Merzbacher Ch 9 (vector spaces, bras/kets, change of basis, Hilbert space) + Ch 10 §10.1–10.5 (eigenvalue problem, commuting observables, uncertainty relations).

Problems (~1.3 hr): **One** change-of-basis problem (position $\leftrightarrow$ momentum) + **one** commuting-observable problem (find simultaneous eigenstates).

Focus: This is the formal heart of QM. If time is tight, skip Ch 9 problem sections and just read.

Week 8 (Jul 20–26) · Angular momentum

Read (~2 hr): Merzbacher Ch 10 §10.6–10.7 (SHO revisited + coherent states — skim) + Ch 11 §11.1–11.4 (orbital ang. mom., algebra, L^2 and L_z , spherical harmonics).

Problems (~1.3 hr): One algebraic angular-momentum problem (raising/lowering between $|l, m\rangle$ states) + one spherical harmonic problem (compute $\langle Y_{lm} | \cos\theta | Y_{lm} \rangle$ or similar).

Focus: The algebra generalises — this is the template for spin and for SU(2) later.

★ Week 9 (Jul 27–Aug 2) — priority · 3D central potentials — hydrogen and spherical wells

Read (~2 hr): Merzbacher Ch 12 in full — reduction of central force, free particle as central-force problem, spherical square well, radial equation and boundary conditions, Coulomb potential, bound-state eigenfunctions for Coulomb. Re-read §12.3 (spherical well) and §12.5–6 (Coulomb) twice.

Problems (~1.3 hr): Three problems (+1 over baseline): (a) spherical square well — find bound states of a given depth and radius (3D version of Ch 6's 1D well); (b) hydrogen expectation values — compute $\langle r \rangle$, $\langle 1/r \rangle$, $\langle 1/r^2 \rangle$ in the ground and $n=2$ states; (c) isotropic 3D harmonic oscillator — get the spectrum by both Cartesian factorisation and the radial equation, then confirm the degeneracies match.

Focus: Priority week. You flagged 3D potential wells as weak. Goal: reduce any central problem to the radial equation in under 5 minutes, then argue the spectrum structure from boundary conditions.

Week 10 (Aug 3–9) · Scattering

Read (~2 hr): Merzbacher Ch 13 §13.1–13.5 — cross sections, wave-packet scattering, Green's functions in scattering, Born approximation, partial waves.

Problems (~1.3 hr): One Born-approximation problem (scattering off a Yukawa or screened Coulomb potential) + one partial-wave / phase-shift problem (hard sphere at low energy).

Focus: Low-energy s-wave is the scattering-test starter — know the sign conventions cold.

★ Week 11 (Aug 10–16) — priority · Spin

Read (~2 hr): Merzbacher Ch 16 §16.1–16.6 — intrinsic angular momentum, QM description of spin, spin and rotations, Pauli matrices, dynamics of a spin system, density matrix for spin. Supplement with Griffiths Ch 4 §4.4 for a gentler first pass if Merzbacher is too compressed.

Problems (~1.3 hr): Three problems (+1 over baseline): (a) Pauli matrix identities — prove $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$ and $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$, then derive $\exp(i\theta \sigma \cdot \mathbf{n}/2)$; (b) spin-1/2 in a magnetic field — Larmor precession and Rabi oscillations for a time-dependent B ; (c) Bloch-sphere problem — start at a given $|\psi\rangle$, apply a sequence of rotations, find the final state.

Focus: Priority week. You flagged spin as weak. Pauli matrix algebra should become reflexive; if it doesn't feel automatic by the end of the week, repeat problem (a) for week 12.

Week 12 (Aug 17–23) · Symmetries + bound-state perturbation + path-integral intro

Read (~2 hr): Merzbacher Ch 17 §17.1–17.5 (rotations, ang. mom. addition, Clebsch–Gordan) + Ch 18 §18.1–18.4 (non-degenerate and degenerate perturbation, polarisability). **Stretch (30 min if time allows):** Merzbacher §15.3 — Feynman's path-integral formulation. Read only; no problems.

Problems (~1.3 hr): One Clebsch–Gordan problem (combine $j_1 = 1$ and $j_2 = 1/2$ explicitly) + one perturbation-theory problem (Stark effect in hydrogen or anharmonic oscillator to 2nd order).

Focus: Stark and Zeeman are must-know applications — understand the logic, not just the answer. The §15.3 path-integral read is a stretch goal: it gives you the idea, not the facility. Treat this as “vocabulary for next year,” not as review.

Week 13 (Aug 24–30) · Time-dependent perturbation + synthesis

Read (~2 hr): Merzbacher Ch 19 §19.1–19.5 — interaction picture, perturbation method, Coulomb excitation, atom in a radiation field, absorption cross section.

Problems (~1.3 hr): One Fermi's golden rule problem (transition rate under a harmonic perturbation) + **one** induced-absorption / selection-rule problem.

Focus: Closing goal: a 1-page cheat sheet for (a) commutator identities, (b) hydrogen spectrum, (c) addition of angular momentum, (d) pert-theory formulas.

Track 3 — Electrodynamics (Zangwill)

Zangwill is long. This plan is curated to cover the qual-standard topics in 13×3.3 hours ≈ 43 hours of work. Whole chapters (12, 13, 18, 19, 23, 24) are deferred for next year. Use Jackson (in reserve) only if Zangwill is insufficient on a specific technique.

Week 1 (Jun 1–7) · Math prelims + Maxwell intro

Read (~2 hr): Zangwill Ch 1 §1.1–1.6 — vectors, derivatives, integrals, generalised functions, Fourier, orthogonal transformations. Ch 2 in full (Maxwell equations in vacuum and in matter).

Problems (~1.3 hr): **One** delta-function / Green's theorem problem from Ch 1 + **one** Ch 2 unit-conversion or Maxwell-equation derivation problem.

Focus: Pin down SI vs. Gaussian conventions now. You'll thank yourself in Ch 22.

Week 2 (Jun 8–14) · Electrostatics core

Read (~2 hr): Zangwill Ch 3 §3.1–3.7 — Coulomb, scalar potential, Gauss' law and solid angle, electrostatic energy (potential and total), stress tensor.

Problems (~1.3 hr): **One** Gauss-law problem with nontrivial symmetry (e.g., non-uniform cylindrical or spherical distribution) + **one** energy-calculation problem (self-energy or work).

Focus: Stress tensor (§3.7) is the entry point to the conservation-law language in Ch 15.

Week 3 (Jun 15–21) · Electric multipoles

Read (~2 hr): Zangwill Ch 4 §4.1–4.6 — dipole, dipole layers, quadrupole, spherical mathematics, spherical/azimuthal multipoles.

Problems (~1.3 hr): **One** multipole-expansion problem (compute the multipole moments of a given charge distribution) + **one** dipole-in-external-field problem (force, torque, energy).

Focus: Know when to use Cartesian vs. spherical multipoles. Skim §4.7 (traceless forms).

Week 4 (Jun 22–28) · Laplace's equation, separation of variables

Read (~2 hr): Zangwill Ch 7 §7.1–7.7 — potential theory, uniqueness, separation in Cartesian/azimuthal/spherical/cylindrical.

Problems (~1.3 hr): **One** Cartesian BVP (charged plates with fixed potentials) + **one** spherical BVP (grounded sphere in a uniform field or similar).

Focus: Boundary-value problems are the E&M half of the qual. Do these two problems cold.

Week 5 (Jun 29–Jul 5) · Poisson's eq., images, Green functions

Read (~2 hr): Zangwill Ch 8 §8.1–8.5 — superposition, method of images, Green-function method, Dirichlet Green function.

Problems (~1.3 hr): **One** classic image problem (point charge near a grounded plane or sphere) + **one** Green-function problem (construct the Dirichlet G for a simple geometry).

Focus: The Green-function argument is the most general BVP tool — don't just memorise images.

Week 6 (Jul 6–12) · Magnetostatics

Read (~2 hr): Zangwill Ch 10 §10.1–10.5 — Biot–Savart, Ampère's law, magnetic scalar/vector potentials.

Problems (~1.3 hr): **One** Ampèrian-loop problem with non-trivial symmetry (solenoid with varying current density, or toroid) + **one** Biot–Savart integration problem.

Focus: Build the habit of writing $A(r)$ first, then $B = \nabla \times A$ — it's cleaner than Biot–Savart by hand.

Week 7 (Jul 13–19) · Magnetic multipoles + forces

Read (~2 hr): Zangwill Ch 11 §11.1–11.3 (dipole, dipole layers) + Ch 12 §12.1–12.6 (charged particle motion, force between currents, magnetic dipole, stress tensor, energy).

Problems (~1.3 hr): **One** charged-particle-in-field problem (cyclotron motion or magnetic bottle) + **one** force/energy problem (torque on a dipole, or force between two current loops).

Focus: Energy of a magnetic system is subtle — §12.6 is worth a slow read.

Week 8 (Jul 20–26) · Dynamic and quasistatic fields

Read (~2 hr): Zangwill Ch 14 §14.1–14.10 — Ampère–Maxwell, Faraday, induction, slowly varying sources, skin effect.

Problems (~1.3 hr): **One** Faraday-law problem (EMF in a moving loop or changing flux) + **one** skin-depth problem (fields in a good conductor at a given frequency).

Focus: Skin effect is a qual classic — know how to get $\delta = \sqrt{2/\mu\sigma\omega}$ in two lines.

Week 9 (Jul 27–Aug 2) · Conservation laws + waves in vacuum

Read (~2 hr): Zangwill Ch 15 §15.1–15.5 (potentials, energy, linear/angular momentum of EM field) + Ch 16 §16.1–16.4 (wave equation, plane waves, polarisation).

Problems (~1.3 hr): **One** Poynting-vector / energy-conservation problem + **one** polarisation problem (circular vs. linear, Stokes parameters).

Focus: Conservation laws are the structural scaffold — Jackson is also good here if needed.

Week 10 (Aug 3–9) · Waves in matter, reflection and refraction

Read (~2 hr): Zangwill Ch 17 §17.1–17.4 — plane waves in simple matter, R&T at interfaces, radiation pressure, layered matter.

Problems (~1.3 hr): **One** Fresnel problem (oblique incidence, find R, T for s- and p-polarisation) + **one** radiation-pressure problem.

Focus: Brewster's angle, total internal reflection — know both cold.

Week 11 (Aug 10–16) · Retarded potentials, dipole radiation

Read (~2 hr): Zangwill Ch 20 §20.1–20.5 — inhomogeneous wave eq., retardation, time-dependent electric dipole, general radiation formula.

Problems (~1.3 hr): **One** Larmor-type radiation problem (accelerating charge in a simple trajectory) + **one** dipole-radiation problem (antenna or atomic transition estimate).

Focus: Retardation is where E&M finally gets hard. Work slowly.

Week 12 (Aug 17–23) · Multipole radiation + scattering

Read (~2 hr): Zangwill Ch 20 §20.6–20.8 (thin-wire antennas, Cartesian multipole, spherical multipole) + Ch 21 §21.1–21.4 (cross sections, Thomson, Rayleigh).

Problems (~1.3 hr): **One** antenna/multipole radiation problem (angular distribution or total power) + **one** Thomson or Rayleigh scattering problem (cross-section derivation).

Focus: Rayleigh scattering's ω^4 is a qual favourite — know why the sky is blue in one sentence.

Week 13 (Aug 24–30) · Special relativity in E&M

Read (~2 hr): Zangwill Ch 22 §22.1–22.7 — Galileo vs. Einstein, Lorentz transformation, four-vectors, electromagnetic quantities, covariant electrodynamics.

Problems (~1.3 hr): **One** four-vector / Lorentz-transformation problem (E and B transformations between frames) + **one** field-tensor problem (verify $F^{\mu\nu}$ form of Maxwell).

Focus: Close the summer by writing Maxwell in covariant form in under 5 minutes from scratch.

What this plan deliberately leaves out

A 130-hour summer cannot cover everything in a first-year syllabus. The following were cut on purpose — either because they are deep topics better served by coursework, or because they're rare on first-year quals:

Track	Deferred topic	Why deferred
CM	Hamilton–Jacobi (Lemos Ch 9)	Conceptually heavy; low yield for review
CM	Action-angle variables, KAM (Ch 9–10)	Advanced / optional in most programs
CM	Classical field theory (Ch 11)	Better encountered in QFT
CM	Relativistic mechanics (Ch 6)	Reappears in E&M Week 13
SM	Phase transitions beyond §11.3	Deep topic; full chapter next term
SM	Non-equilibrium, transport (Pathria 15)	Second-semester SM
SM	Monte Carlo / MD (Pathria 16)	Computational; separate track
QM	Path integrals (Ch 15, full)	Brief intro in wk 12 stretch; deep treatment in advanced QM
QM	2nd quantisation, many-body (Ch 21–22)	Advanced QM course
QM	Relativistic (Dirac) — Ch 24	Covered in QFT
E&M	Dielectric and magnetic matter (Ch 6, 13)	Light touch via Ch 2 is enough
E&M	Waveguides (Ch 19)	Practical; revisit if qual emphasises
E&M	Dispersive media (Ch 18)	Advanced E&M
E&M	Synchrotron/Cherenkov (Ch 23)	Specialised; covered next year

Staying on pace

Three rules, in priority order:

- 1. If you fall behind, drop reading before you drop problems.** Problem-solving is the retrieval practice that makes review stick; reading without problems mostly fades. A week where you read 10 pages and did 6 problems beats a week where you read 30 and did 2.
- 2. Protect one full hour of continuous work per track per week.** Three 20-minute sessions are not equivalent to one 60-minute session. Hard problems require ramp-up time.
- 3. Every 4 weeks, spend 30 minutes re-scanning the old weeks.** Read only the margin notes and starred problem solutions. Anything you cannot recall within 30 seconds goes on a 'revisit' list for the final week.

If you finish everything early

Unlikely at 10 hr/week — but possible. In priority order: (a) go back and add Goldstein Ch 10 (Hamilton–Jacobi) and Lemos Ch 9 for CM depth, (b) add Merzbacher Ch 14–15 for dynamics / path integrals, (c) add Zangwill Ch 19 (waveguides) and Ch 18 (dispersion). Do not add more books — going deep beats going wide.

If you fall a full week behind

Week 13 is a synthesis / soft-finish week in QM and E&M. In SM, week 13 is new material (Brownian motion) — so if you fall behind on SM, catch up in the first week of September rather than shortening week 13.

A note on path integrals and CFT

You flagged these as things you have never done. They are worth naming clearly: they are **new learning**, not review. Treating them as review items would either force another topic off the plan or stretch your weekly budget past 10 hours. Two recommendations:

Path integrals. Merzbacher §15.3 gives a competent first exposure (~8 pages). Read it in week 12 as a stretch goal. For real fluency, plan a 2–3 week dedicated study in the fall using Shankar Ch 8 or Feynman & Hibbs — this is its own project, not a review item.

CFT. This term is ambiguous. If you mean **classical field theory** (Lagrangian density, Noether currents, classical EM as a field theory) — this is the natural next step after CM and is covered in Lemos Ch 11. It is deferred here but worth doing as a ~4-hour block after the summer plan ends. If you mean **conformal field theory** or **quantum field theory** — that is multi-semester advanced coursework and does not belong in a first-year review at all. (QFT typically comes after a full year of graduate QM, E&M, and usually a semester of classical field theory.) Tell me which one you meant and I'll either (a) add a Lemos Ch 11 block as a week-14 extension, or (b) sketch a longer-range CFT/QFT roadmap separately.